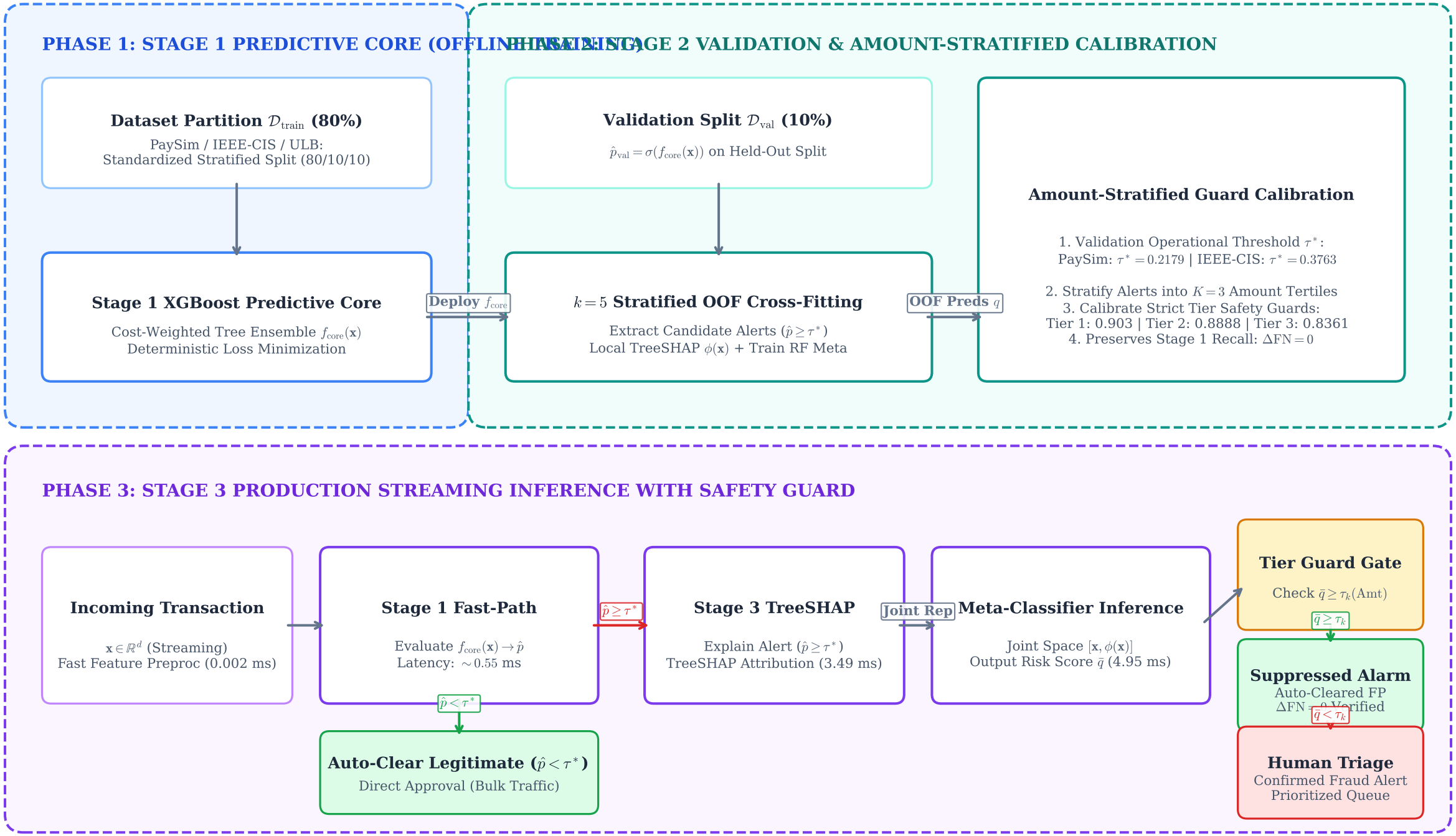

PHASE 1: STAGE 1 PREDICTIVE CORE (OFFLINE TRAINING)
Dataset Partition $\mathcal{D}_{\text{train}}$ (80%)
PaySim / IEEE-CIS / ULB:
Standardized Stratified Split (80/10/10)
Stage 1 XGBoost Predictive Core
Cost-Weighted Tree Ensemble $f_{\text{core}}(\mathbf{x})$
Deterministic Loss Minimization
Deploy f_{core}
PHASE 2: STAGE 2 VALIDATION & AMOUNT-STRATIFIED CALIBRATION
Validation Split $\mathcal{D}_{\text{val}}$ (10%)
$\hat{p}_{\text{val}} = \sigma(f_{\text{core}}(\mathbf{x}))$ on Held-Out Split
$k=5$ Stratified OOF Cross-Fitting
Extract Candidate Alerts ($\hat{p} \geq \tau^*$)
Local TreeSHAP $\phi(\mathbf{x})$ + Train RF Meta
OOF Preds q
Amount-Stratified Guard Calibration
1. Validation Operational Threshold τ^*:
PaySim: $\tau^* = 0.2179$ | IEEE-CIS: $\tau^* = 0.3763$
2. Stratify Alerts into $K=3$ Amount Tertiles
3. Calibrate Strict Tier Safety Guards:
Tier 1: 0.903 | Tier 2: 0.8888 | Tier 3: 0.8361
4. Preserves Stage 1 Recall: $\Delta\text{FN} = 0$
PHASE 3: STAGE 3 PRODUCTION STREAMING INFERENCE WITH SAFETY GUARD
Incoming Transaction
$\mathbf{x} \in \mathbb{R}^d$ (Streaming)
Fast Feature Preproc (0.002 ms)
Stage 1 Fast-Path
Evaluate $f_{\text{core}}(\mathbf{x}) \to \hat{p}$
Latency: ~ 0.55 ms
$\hat{p} \geq \tau^*$
$\hat{p} < \tau^*$
Auto-Clear Legitimate ($\hat{p} < \tau^*$)
Direct Approval (Bulk Traffic)
Stage 3 TreeSHAP
Explain Alert ($\hat{p} \geq \tau^*$)
TreeSHAP Attribution (3.49 ms)
Joint Rep
Meta-Classifier Inference
Joint Space $[\mathbf{x}, \phi(\mathbf{x})]$
Output Risk Score $\bar{q}$ (4.95 ms)
Tier Guard Gate
Check $\bar{q} \geq \tau_k(\text{Amt})$
$\bar{q} \geq \tau_k$
Suppressed Alarm
Auto-Cleared FP
$\Delta\text{FN} = 0$ Verified
$\bar{q} < \tau_k$
Human Triage
Confirmed Fraud Alert
Prioritized Queue